

01 / Arrays and matrix basics

First recognise scalars, vectors and matrices. Then learn the basic operations.

A. Mathematical objects

Scalar

A single numerical value.

$$s = 7$$

Shape: a single value.

Row vector

One row of values.

$$\mathbf{r} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

Shape: 1×3 .

Column vector

One column of values.

$$\mathbf{c} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Shape: 3×1 .

Matrix (rectangular array)

Numbers arranged in rows and columns.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Shape: 2×3 .

Remember: matrix shape is always **rows** $\times$ **columns**. For $\mathbf{A} \in \mathbb{R}^{m \times n}$, the entry A_{ij} is at **row** i , **column** j . Vectors are matrices with exactly one row or column.

B. Basic operations

Addition and subtraction

Only matrices of the **same shape** can be added. Combine entries in the same position:

$$(\mathbf{A} + \mathbf{B})_{ij} = A_{ij} + B_{ij}, \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}.$$

Multiplication by a scalar

Multiply **every entry** by the number λ ; the shape is unchanged.

$$(\lambda \mathbf{A})_{ij} = \lambda A_{ij}, \quad 2 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}.$$

Transpose $\mathbf{A}^\top$

Turn rows into columns. An $m \times n$ matrix becomes $n \times m$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^\top = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}, \quad (\mathbf{A}^\top)_{ij} = A_{ji}.$$

02 / Matrix algebra

Matrix multiplication, useful identities, norms and inverses.

A. Multiplying matrices

Check dimensions first

The **inner dimensions must match**; the outer dimensions give the result:

$$\underbrace{A}_{m \times n} \underbrace{B}_{n \times p} = \underbrace{C}_{m \times p}, \quad C_{ik} = \sum_{j=1}^n A_{ij} B_{jk}.$$

Each output entry is one row of A multiplied entry-by-entry with one column of B , then summed.

Worked example: row $\times$ column

$$\underbrace{\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}}_A \underbrace{\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}}_B = \underbrace{\begin{bmatrix} 23 & 34 \\ 31 & 46 \end{bmatrix}}_{AB}$$

For the first entry, take **row 1** of A and **column 1** of B :

$$(AB)_{11} = 5 \cdot 1 + 6 \cdot 3 = 23.$$

Order matters: in general $AB \neq BA$. Sometimes one order is not even defined.

B. Rules and special matrices

Two multiplication identities

$$(AB)C = A(BC) \quad (AB)^T = B^T A^T.$$

Multiplication is associative. Transposing a product reverses factor order.

Euclidean norm (length)

For a column vector $\mathbf{v}$:

$$\|\mathbf{v}\|_2 = \sqrt{\sum_i v_i^2}$$

$$\|\mathbf{v}\|_2^2 = \mathbf{v}^T \mathbf{v}.$$

Identity matrix I_n

Diagonal entries are 1, the others 0. It leaves a compatible matrix unchanged:

$$I_m A = A I_n = A.$$

C. Inverses and solving systems

A square matrix A is **invertible** if $AA^{-1} = A^{-1}A = I$ for some inverse A^{-1} . For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the 2×2 inverse exists exactly when $ad - bc \neq 0$:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

If A is invertible, the unique solution of $A\mathbf{x} = \mathbf{b}$ is $\mathbf{x} = A^{-1}\mathbf{b}$. A square matrix with no inverse is called **singular**.

03 / Differential calculus

Derivatives measure how quickly a function changes.

A. What is a derivative?

$f'(x)$ is the **instantaneous rate of change**: the slope of the tangent to the graph.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Example: $f(x) = x^2 \implies f'(x) = 2x \implies f'(1) = 2$.

B. Four derivatives to remember

Constant

$$\frac{d}{dx} c = 0$$

A constant does not change.

Power

$$\frac{d}{dx} x^p = p x^{p-1}$$

Bring the exponent down, then subtract 1.

Exponential

$$\frac{d}{dx} e^x = e^x$$

The derivative is unchanged.

Natural logarithm

$$\frac{d}{dx} \log x = \frac{1}{x} \quad (x > 0)$$

The real logarithm requires $x > 0$.

C. Rules for combined functions

Linearity

Differentiate each term separately and keep its numerical coefficient.

$$[au(x) + bv(x)]' = a u'(x) + b v'(x).$$

Product rule

Differentiate each factor in turn while keeping the other factor unchanged.

$$(uv)' = u'v + uv', \quad [x^2(x+1)]' = 2x(x+1) + x^2.$$

Chain rule: outer derivative $\times$ inner derivative

For a composition $F(s) = v(u(s))$:

$$F'(s) = v'(u(s)) u'(s).$$

Worked example:

$$\frac{d}{ds} (2s+5)^3 = 3(2s+5)^2 \cdot 2 = 6(2s+5)^2.$$

04 / Gradients and minima

Differentiate functions of several variables, then locate and check minima.

A. Partial derivatives and gradients

A **partial derivative** varies one input and holds all other inputs fixed. For example:

$$F(a, b) = a^2 + 3ab + b^2,$$

$$\frac{\partial F}{\partial a} = 2a + 3b, \quad \frac{\partial F}{\partial b} = 3a + 2b.$$

The **gradient** stacks the partial derivatives into a column vector:

$$\nabla F(a, b) = \begin{bmatrix} \partial F / \partial a \\ \partial F / \partial b \end{bmatrix} = \begin{bmatrix} 2a + 3b \\ 3a + 2b \end{bmatrix}.$$

If a function is differentiable, the gradient gives its linear approximation. Having partial derivatives alone does not always guarantee differentiability.

Partial chain rule

Apply the ordinary chain rule to the selected variable; hold the others fixed. For $R(a, b) = (a + 2b - 3)^2$:

$$\frac{\partial R}{\partial a} = 2(a + 2b - 3), \quad \frac{\partial R}{\partial b} = 4(a + 2b - 3).$$

B. Finding minima

Local minimum

Smallest value compared with **nearby** inputs.

Global minimum

Smallest value over the **entire domain**.

Fermat's theorem: a necessary condition

At a differentiable **interior local minimum**, the derivative (or gradient) is zero:

$$\boxed{f'(x^*) = 0} \quad \text{or} \quad \boxed{\nabla H(u^*) = 0}.$$

A point where the derivative/gradient vanishes is a **stationary point**. This condition alone does **not** prove a minimum. **Check** each candidate against the function and its domain.

C. Stationary does not mean minimum

$$\begin{aligned} f(x) = x^2 &\Rightarrow \text{minimum at } 0, \\ f(x) = -x^2 &\Rightarrow \text{maximum at } 0, \\ f(x) = x^3 &\Rightarrow \text{stationary inflection at } 0. \end{aligned}$$

In several variables, a stationary point may be a **saddle**. Example:

$$S(u_1, u_2) = u_1^2 - u_2^2, \quad \nabla S(0, 0) = 0,$$

which increases along one axis and decreases along the other: the origin is *not* a minimum.